

Phase 1: Ingest

Getting one company's raw SEC filings onto disk, reliably and cheaply re-usable.

What this phase does

Every US public company's financial filings are available, for free, as structured data: the SEC's XBRL "companyfacts" API returns one JSON document per company containing every number that company has ever tagged in any filing, going back years. Phase 1's whole job is narrow: resolve a ticker to a CIK (SEC's company identifier), fetch that JSON, and cache it to disk so nothing downstream ever needs the network again.

```
GET https://data.sec.gov/api/xbrl/companyfacts/CIK0000320193.json
```

```
-> { "entityName": "Apple Inc.",  
    "facts": { "us-gaap": {  
      "Assets": { "units": { "USD": [  
        {"end": "2023-09-30", "val": 352583000000,  
        "fy": 2023, "fp": "FY", "form": "10-K", ...}  
      ] } }, ... } } }
```

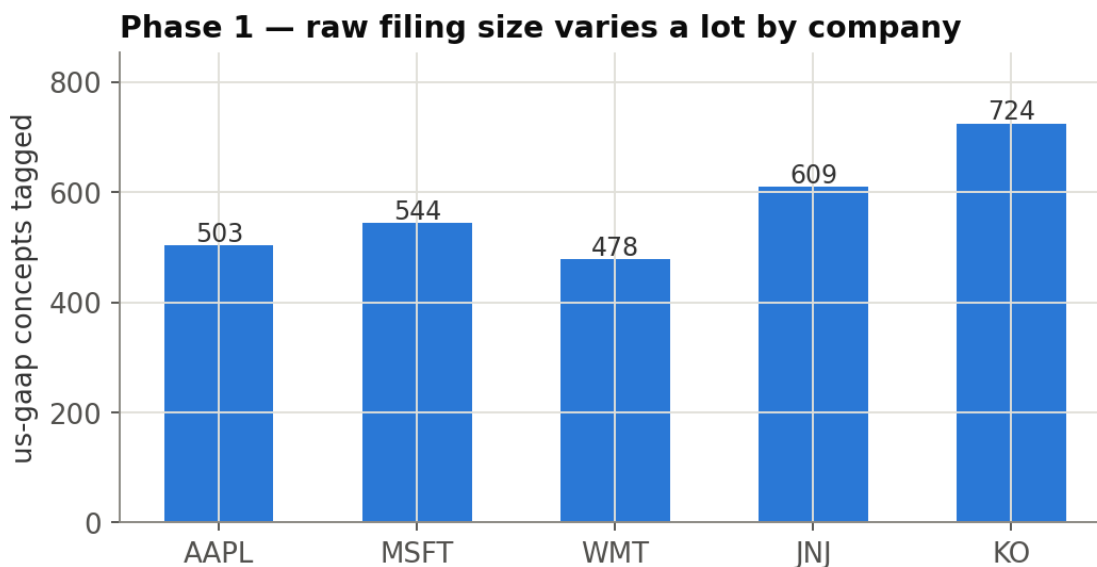
The core idea: cache everything, hit the network once

SEC explicitly asks for a descriptive User-Agent with contact information and reasonable request rates -- this is a shared public resource, not a private API. The ingest layer sends that User-Agent on every request, retries transient failures (429s, 5xx) with exponential backoff, and writes every response to data/cache/, keyed by CIK. Every later phase reads through this same cache. The practical effect: you can iterate on normalization logic, run the tool a hundred times, and never re-download a byte from SEC after the first fetch.

What real data looks like before you've done anything with it

Even at the raw-ingest stage, one thing is already visible: companies don't tag the same amount of detail. The number of distinct us-gaap concepts a company has ever used varies by industry and filing history -- more concepts doesn't mean a bigger company, it means a more complex or more thoroughly-tagged one.

STATEMENT ANALYZER — PHASE 1



us-gaap concepts tagged, by company (from real cached companyfacts JSON).

Company	CIK	us-gaap concepts	Fiscal years cached
Apple Inc.	0000320193	503	17
Microsoft Corp.	0000789019	544	16
Walmart Inc.	0000104169	478	17
Johnson & Johnson	0000200406	609	16
Coca-Cola Co.	0000021344	724	17

KEY TAKEAWAY

Phase 1 deliberately does no interpretation at all -- it doesn't know what "Assets" means, it just fetches and caches. That discipline matters: when something looks wrong three phases later, you can always go back to the exact raw JSON on disk and know it hasn't been silently transformed anywhere upstream.